Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

2.REQUIREMENT ANALYSIS

2.1. DATA FLOW DIAGRAM & USER STORIES

2.1.1 User Story

The **Automating Employee Data Import System** is designed to simplify the process of importing employee data into the ServiceNow platform and automatically displaying relevant employee and managerial information during incident management.

In many organizations, employee data is stored in external systems such as spreadsheets or HR databases. Service desk agents often need employee information when assigning incidents. Manually entering or searching for this data is time-consuming and prone to errors. The proposed system automates the process of importing employee records and integrates them directly into ServiceNow.

The system enables administrators to import employee data such as name, email, department, manager name, and manager email from external files like CSV or Excel. Once the data is imported, it is automatically stored in the ServiceNow user database. When incidents are assigned to employees, the system automatically retrieves related information such as department details and manager email using dot walking.

This automation significantly reduces manual work for administrators and improves the efficiency of incident handling. Service desk agents can quickly view the department and manager details of the assigned employee directly on the incident form. This ensures faster communication and escalation when required.

Additionally, incidents typically require faster response times than other tasks. To address this, the system modifies the default priority value of the incident table using dictionary override. This ensures that incidents are assigned an appropriate priority level automatically.

By integrating automated data import, incident management, and customized priority settings, the system provides a streamlined workflow that improves productivity, data accuracy, and response time for IT support teams.

2.1.2 Data Flow Diagram (DFD)

A **Data Flow Diagram (DFD)** illustrates how data moves within the system and how different components interact to process and store information.

EXTERNAL ENTITY – EMPLOYEE DATA SOURCE

The process begins with an external data source such as a spreadsheet or HR system containing employee details. The data typically includes fields like employee name, email, department, manager name, and manager email. This data is exported into formats such as CSV, XML, or XLSX before being uploaded to the ServiceNow platform.

Process 1 – Import Set

Once the file is uploaded, ServiceNow creates an **Import Set Table**. This table acts as a temporary storage location where the imported data is stored before being processed. The import set ensures that the data can be validated and transformed before it is inserted into the main database.

Process 2 – Transform Map

After the data is loaded into the import set table, a **Transform Map** is used to map the fields from the staging table to the corresponding fields in the target table. This mapping ensures that employee attributes such as name, department, and manager information are correctly transferred to the ServiceNow user table.

The transform map also performs data validation and transformation during this process. Any mismatched or incorrect data can be identified and corrected before being inserted into the system.

Process 3 – User Table Storage

Once the transformation is completed, the employee data is inserted into the **ServiceNow User Table (sys_user)**. This table stores all user records within the platform and allows other modules such as incident management to access employee information.

Process 4 – Incident Management

When an incident is created in the system and assigned to a specific employee, the incident form retrieves relevant employee details automatically. Using dot walking, the system can display related fields such as department, employee email, and manager email directly on the incident form.

This ensures that service desk agents have immediate access to essential information required for incident resolution and escalation.

Process 5 – Dictionary Override

The incident table is derived from the task table and inherits its default properties. However, incidents typically require a different default priority level. Using the dictionary override feature, the system modifies the default priority value specifically for incidents.

This ensures that incidents receive a priority level that reflects their urgency without affecting the parent task table.